

E-Commerce Business Analysis Report

1. Executive Summary

This report presents a comprehensive analysis of an E-commerce transactional dataset with the objective of deriving meaningful business insights. The dataset consists of 541,909 records, which were cleaned to 397,884 valid transactions after removing missing and inconsistent data such as null customer IDs and negative values in quantity and price.

The analysis reveals that the business generated a total revenue of approximately 8.9 million, with a strong concentration in the United Kingdom, which contributes the majority of sales. Additionally, a clear seasonal trend is observed, with peak sales occurring in November, likely due to holiday shopping behaviour.

The report highlights key performance drivers, including top-selling products, geographic concentration, and time-based trends. Based on the findings, strategic recommendations are provided to improve business performance and scalability.

2. Objectives of the Analysis

The primary objectives of this analysis are:

- To understand the structure and quality of the dataset
- To identify key revenue drivers
- To analyze product performance and demand patterns
- To examine geographic sales distribution
- To detect seasonal trends in sales
- To provide actionable business recommendations

3. Dataset Overview

The dataset represents transaction-level data from an online retail business. Each row corresponds to a product purchased within a transaction.

Key Attributes:

- InvoiceNo – Unique transaction ID
- StockCode – Product ID
- Description – Product name
- Quantity – Number of units purchased
- InvoiceDate – Date of transaction
- UnitPrice – Price per unit
- CustomerID – Unique customer identifier
- Country – Customer location

4. Data Cleaning and Preprocessing

Before analysis, the dataset underwent significant cleaning to ensure accuracy and reliability.

Issues Identified:

- Missing values in CustomerID (135,080 rows)
- Missing values in Description (1,454 rows)
- Negative values in Quantity (indicating returns)
- Negative values in UnitPrice (invalid entries)

Actions Taken:

- Removed rows with missing CustomerID
- Filtered out transactions with negative Quantity
- Removed entries with negative UnitPrice
- Converted InvoiceDate into datetime format

Result:

- ✓ Clean dataset reduced to 397,884 records
- ✓ Data is now consistent and suitable for analysis

5. Revenue Analysis

Total Revenue:

- **8,911,407.904**

This indicates strong business performance over the given period.

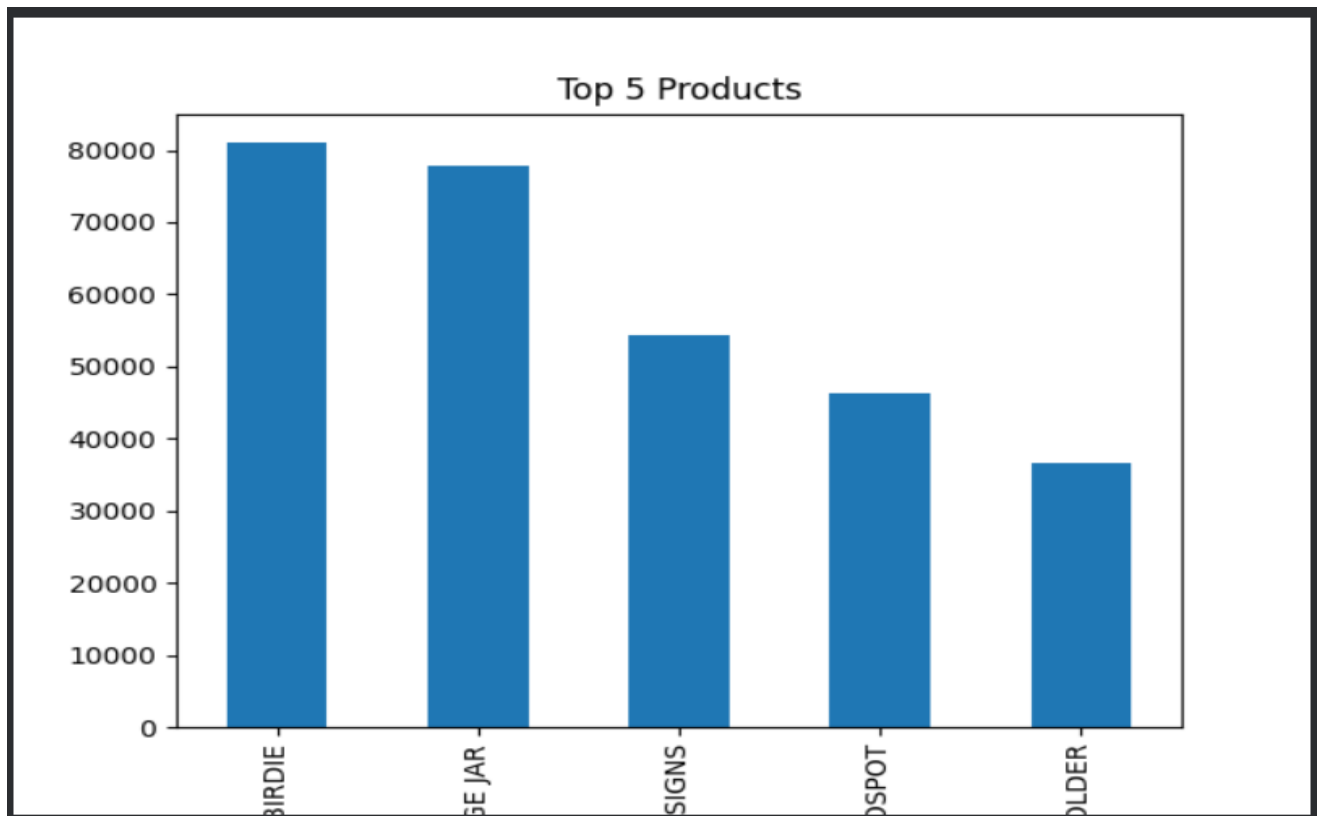
Key Observations:

- Revenue is significantly high, indicating strong sales volume
- Majority of revenue comes from a limited set of products and regions

6. Product Performance Analysis

Top 5 Selling Products (by Quantity):

- PAPER CRAFT, LITTLE BIRDIE
- MEDIUM CERAMIC TOP STORAGE JAR
- WORLD WAR 2 GLIDERS ASSTD DESIGNS
- JUMBO BAG RED RETROSPOT
- WHITE HANGING HEART T-LIGHT HOLDER



Insights:

- These products contribute significantly to total sales
- Most are decorative or gift-related items
- High demand suggests strong customer preference pattern

Business Implication:

- These products should be prioritized in inventory management
- Can be used for promotional campaigns and bundling strategies

7. Geographic Analysis

Top 5 Countries by Revenue:

1. United Kingdom
2. Netherlands
3. EIRE
4. Germany
5. France

Insights:

- The **United Kingdom dominates sales**, contributing the majority of revenue
- Other countries contribute significantly less

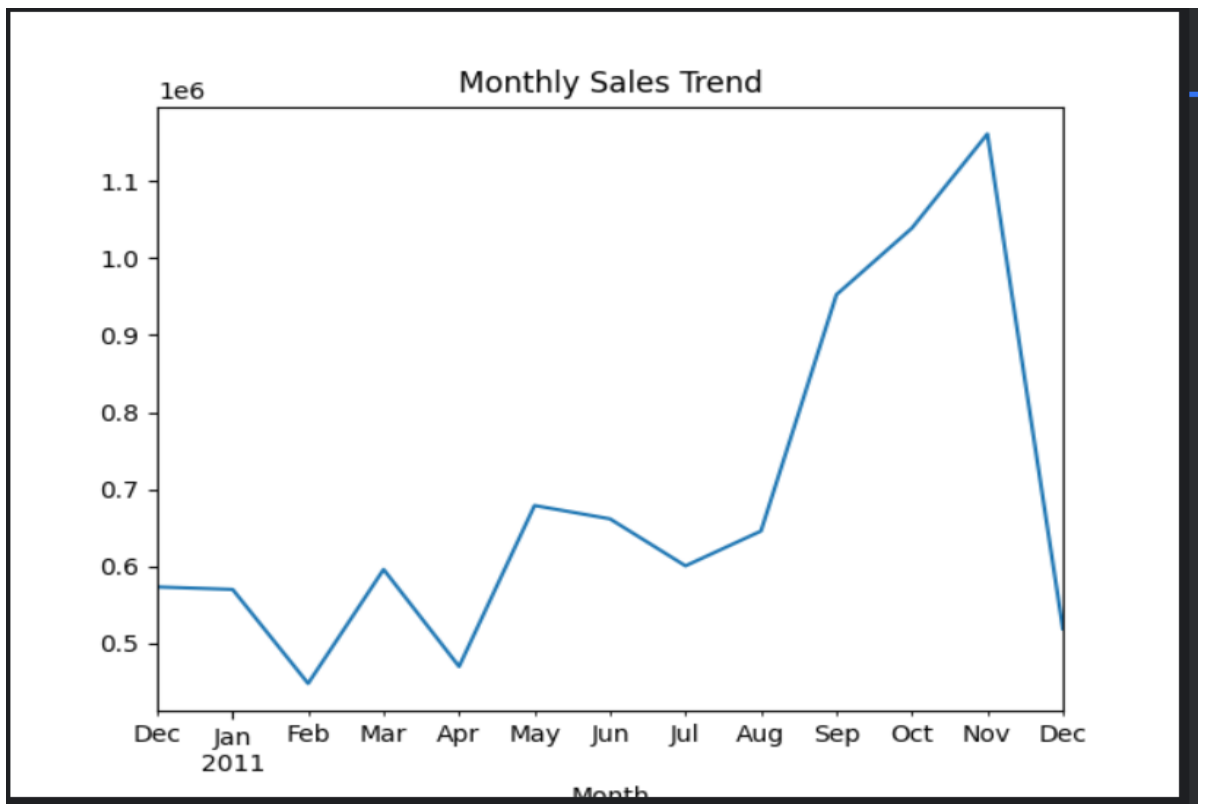
Business Implication:

- Heavy dependency on a single market (UK)
- Risk of revenue concentration
- Opportunity to expand into other regions

8. Time-Based (Seasonal) Analysis

Monthly Trend Observations:

- Sales increase gradually from mid-year
- Peak observed in **November**
- Lower sales in early months like February and April



Insights:

- Strong **seasonal demand pattern**
- Likely influenced by:
 - Holiday season
 - Year-end shopping behavior

Business Implication:

- Increase inventory before peak months
- Plan marketing campaigns around seasonal demand
- Optimize supply chain for high-demand periods

9. Key Business Insights

- The business is generating **high revenue (~8.9M)**
- A small number of products drive a large portion of sales
- Revenue is highly concentrated in the **UK market**
- Sales show strong **seasonality**, peaking in November
- Data quality issues highlight the importance of preprocessing

10. Strategic Recommendations

i. Inventory Optimization

- Maintain higher stock levels for top-selling products
- Avoid stockouts during peak demand periods

ii. Market Expansion

- Increase focus on countries like Germany and France
- Develop targeted marketing strategies

iii. Seasonal Planning

- Prepare for high sales during Q4
- Launch promotions before peak months

iv. Data Quality Improvement

- Ensure proper customer data collection
- Reduce missing values for better analysis

v. Customer Analytics (Future Scope)

- Perform segmentation using CustomerID
- Identify high-value customers

11. Limitations of the Analysis

- Removed data may affect total revenue accuracy
- Customer-level insights are limited due to missing data
- Returns and cancellations were excluded

12. Decision Making

Insight	Business Action	Impact
High UK dependency	Expand into international markets	Reduces risk and increases revenue diversification
Peak in November	Increase inventory before Q4	Prevents stockouts and maximizes profit
Top products dominate	Focus marketing on best-selling items	Improves sales efficiency and ROI
Missing customer data	Improve data collection systems	Enables better customer analytics and personalization

13. Conclusion

The analysis demonstrates that the E-commerce business is performing strongly, with high revenue and consistent demand for specific products. However, the business is highly dependent on a single geographic region and exhibits seasonal variability.

By leveraging these insights, the company can improve operational efficiency, expand market reach, and enhance profitability.